VRE Investment Thesis

Is Variable Renewable Energy Development Still Worthwhile? A Data-Center-Demand Analysis

Last updated: March 12, 2026 • Data sources: LBNL, EPRI, S&P Global, PJM, ERCOT, LevelTen, BNEF

The Core Question

Does VRE (solar/wind) development generate attractive returns for suppliers, given accelerating price cannibalization, **without** relying on market reform or carbon pricing? And if so, where, in what form, and for whom?

> **Short answer:** Yes, but conditionally. The corporate PPA market — driven overwhelmingly by hyperscaler sustainability commitments — creates a **willingness-to-pay floor** that sits well above cannibalized wholesale value. This premium is the hedge. But it only works in the right ISOs, with the right product (increasingly storage-paired), and only as long as at least some hyperscalers maintain ambitious climate targets. The investment thesis lives or dies on the durability of corporate demand, not wholesale market fundamentals.

1. The Cannibalization Landscape

Solar and wind capture rates — the ratio of generation-weighted price to flat average price — are declining at different speeds across the 7 US ISOs. This is the core risk for VRE developers: as more zero-marginal-cost generation enters, it suppresses prices during its own generation hours.

Solar Capture Rate Trajectory by ISO (2020–2035 est.)

Sources: LBNL Utility-Scale Solar 2024; S&P Global Commodity Insights; REsurety; author estimates for 2026–2035

ISO	SOLAR CAPTURE 2024	SOLAR CAPTURE 2030E	WIND CAPTURE 2024	STAGE	KEY DRIVER
CAISO	**<30%**	~15%	85%	Advanced	30% solar penetration. 13% negative-price hours. 3.4M MWh curtailed.
ERCOT	**70%**	~45%	54%	Active	15% solar & rising. 8 TWh curtailed in 2024. Prices halved YoY.
PJM	**95%**	~70%	90%	Early	Curtailment jumped 6x in 2024. Solar growing fast. 3–5 year window.
MISO	**98%**	~70%	78%	Early → Active	Solar 4x since 2023 (4.2→16.2 GW). Approaching inflection.
SPP	**98%**	~72%	58%	Solar early; Wind severe	Wind 42% below flat. Worst wind capture in US.
NYISO	**100%**	~88%	96%	Pre-cannibalization	Low penetration. Constrained market. Moratorium risk.
NEISO	**100%**	~90%	95%	Pre-cannibalization	Highest power costs. Offshore wind pipeline.

Spain: the endgame without storage. Spain's solar capture rate fell to 56% in 2025 H1, with 500+ negative-price hours, project valuations down 60%+, and only 18 MW of standalone BESS deployed nationally. This is where CAISO ends up without continued storage deployment — and where ERCOT/PJM/MISO are headed on a 5–10 year lag.

2. The Hyperscaler Demand Hedge

The critical insight: **VRE developers don't need healthy wholesale markets if they have corporate PPA offtakers willing to pay above wholesale.** The "green premium" — the gap between a corporate PPA price and the cannibalized capture value — is the entire margin story.

The Green Premium: PPA Price vs. Capture-Weighted Wholesale Value by ISO (2025)

Sources: LevelTen PPA Price Index Q3 2025; LBNL value factor data; author synthesis

Who's buying, and what do they need?

COMPANY	APPROACH	US CAPACITY (GW)	CLEAN PPAS (GW)	UNMET NEED (GW)	PRODUCT DEMAND
Google	Hourly (24/7 CFE)	8.0	3.5	4.5	Shaped clean: solar+storage, wind, nuclear, geothermal. Same ISO.
Microsoft	Hourly (24/7 CFE)	7.5	4.0	3.5	Shaped clean: nuclear (TMI), solar+storage, wind. Same ISO.
Amazon/ AWS	Consequential	12.0	6.0	6.0	Cheapest VRE anywhere. Wind RECs in MISO/SPP. Volume over location.
Meta	Consequential	6.0	2.0	4.0	Cheapest VRE + gas. Offset strategy. Nuclear for optics.
Equinix	Annual	3.0	1.5	1.5	Green tariffs + PPAs. Annual matching.
Digital Realty	Annual (SBTi)	2.5	1.0	1.5	Mix of PPAs + RECs.
Oracle	None	3.0	0.3	—	No clean procurement mandate.
CoreWeave	None	3.0	0.0	—	Pure compute. No sustainability commitments.
xAI	None	1.0	0.0	—	Gas turbines. No clean commitments.

~20 GW of unmet hyperscaler clean energy demand exists today, growing to 40–60 GW by 2030 as data center load doubles. Google and Microsoft alone need ~8 GW of *new, shaped* clean energy by 2030. Meta and Amazon need ~10 GW of *any* VRE for annual/consequential claims. This is the addressable market for VRE developers.

3. Where Does VRE Investment Make Sense?

VRE Investment Attractiveness by ISO (Combining Cannibalization Risk, Demand, and Premium)

Author analysis combining capture rate projections, hyperscaler demand density, PPA premium, and interconnection feasibility

ISO	PRODUCT	PPA PRICE	CAPTURE VALUE	GREEN PREMIUM	DEVELOPER IRR	RATING	WHY
PJM	Solar+Storage	$85/ MWh	$35	**$50**	9–11%	**Strong Buy**	Largest DC market. 30 GW DC growth. $666/MW-day capacity. Massive unmet demand.
PJM	Solar standalone	$70	$49	$21	7–9%	**Buy (shrinking window)**	Good today but 3–5yr window before capture erodes. Lock in PPAs now.
ERCOT	Solar+Storage	$60	$20	**$40**	8–10%	**Strong Buy**	233 GW load queue. Battery arbitrage $150+/ kW-yr. Behind-meter DC demand.
ERCOT	Solar standalone	$42	$27	$15	6–8%	**Hold**	Cannibalization accelerating. No capacity market floor. Merchant tail risky.
MISO	Solar+Storage	$65	$37	$28	8–10%	**Buy**	Meta/Amazon primary market. Conditional on consequential accounting surviving.
MISO	Wind	$45	$27	$18	6–8%	**Hold**	Already well-supplied. Cheap RECs ($1–5). Diminishing marginal value.
NYISO	Solar+Storage	$95	$66	$29	9–11%	**Buy**	Premium market. No cannibalization. Hard to build (permitting).
NEISO	Solar+Storage	$100	$72	$29	8–10%	**Buy (small)**	Premium but tiny market. Scale-limited.
CAISO	Solar standalone	$75	$5	**$70 (illusory)**	4–6%	**Avoid**	<30% capture. Unbankable standalone. Curtailment risk.
CAISO	Solar+Storage	$90	$25	$65	7–9%	**Buy (selective)**	Must be storage-paired. Storage saturation risk emerging.
SPP	Wind	$40	$8	$32 (fragile)	5–7%	**Avoid**	58% capture. Worst wind market. No hyperscaler anchor.

4. The Accounting Regime Determines Everything

The GHG Protocol Scope 2 revision — expected mid-2026 to end-2027 — is the single biggest determinant of VRE investment value. The outcome dictates *what kind* of clean energy has value, *where* it needs to be built, and *how much* hyperscalers will pay.

Scenario A: Hourly Matching Wins

Scenario B: Consequential Wins

FACTOR	HOURLY MATCHING WINS	CONSEQUENTIAL WINS	HYBRID (CURRENT REALITY)
VRE demand location	Same ISO as data center load	Cheapest possible (MISO/SPP wind)	Split: hourly buyers local, consequential buyers anywhere
VRE product needed	Shaped (solar+storage, wind+storage)	Cheapest unshaped (standalone wind)	Mix of both
Clean firm demand	High (nuclear, geothermal, LDES)	Near zero	Moderate (Google/MSFT only)
PPA premium over wholesale	30–50% (shaped product commands premium)	5–15% (unbundled RECs race to bottom)	15–30%
Storage investment signal	Very strong	Weak	Moderate
Best ISO for VRE developer	PJM, ERCOT, CAISO (where DCs are)	MISO, SPP (cheapest production)	PJM best in all scenarios
MISO wind RECs value	Near zero (wrong ISO for buyers)	High (cheapest $/tCO2 avoided)	Moderate
Total clean energy demand (GW by 2030)	~40 GW	~40 GW	~30 GW
Developer IRR impact	Higher margins, higher capex	Lower margins, lower capex	Mixed

> **PJM solar+storage is the dominant position regardless of accounting outcome.** Under hourly matching, PJM is where Google/Microsoft need shaped clean energy. Under consequential accounting, PJM still has massive DC load and capacity market payments ($666/MW-day). It's the only ISO where VRE investment is robust to all policy scenarios.

5. The Five Risks

Risk 1: Capture Rate Erosion

The most quantifiable risk. Every ISO is on the cannibalization curve — the question is *where*. CAISO is the warning: solar went from 146% capture (2012) to <30% (2024). PJM and MISO are 3–7 years behind on the same trajectory. **Mitigation:** Storage pairing lifts capture ~30%. Lock in 15-year PPAs before rates decline further.

Risk 2: Accounting Regime Uncertainty

If the GHG Protocol mandates hourly matching, MISO/SPP wind RECs become worthless for corporate claims. If consequential wins, shaped products lose their premium. **Mitigation:** Develop in PJM (robust to both). Pair with storage (valuable in either regime). Avoid pure REC plays in MISO.

Risk 3: Hyperscaler Demand Durability

Corporate clean energy buying fell 10% in 2025 (BNEF) after nearly a decade of growth. Meta and Amazon's emissions are rising 34–65% despite pledges. If hyperscalers abandon targets, the green premium disappears. **Mitigation:** Target long-term PPAs (15–20yr). Diversify across buyers. Capacity market revenue provides a floor in PJM/NYISO.

Risk 4: Policy / Tax Credit Erosion

The One Big Beautiful Bill Act cut most IRA clean energy incentives. Solar/wind must begin construction by July 2026 for full ITC/PTC. **Mitigation:** Safe harbor ASAP. Projects already in construction are protected. Battery ITC standalone still available through 2032.

Risk 5: Interconnection Queue Congestion

PJM has 170+ GW in its generation queue. ERCOT has 432 GW. Completion rates are 15–25%. Projects wait 4–6 years for interconnection. **Mitigation:** Co-location with data centers (behind-meter). Acquire queue positions. Target ISOs with better throughput (ERCOT > PJM for queue velocity).

6. The Five Opportunities

Opportunity 1: The Supply-Demand Gap is Enormous. ~20 GW of unmet hyperscaler demand today. Growing to 40–60 GW by 2030. PJM alone needs 30 GW of new generation capacity for data centers. Only 57 GW of projects have even been offered interconnection agreements across the entire PJM footprint. Demand >> supply for years.

Opportunity 2: Shaped Products Command Premium Margins. Solar+4hr storage PPAs at $85/MWh in PJM vs. $35/MWh capture-weighted wholesale value = $50/MWh green premium. That's a 140% premium over wholesale. Battery pairing adds $10–15/MWh to LCOE but $50+ to PPA revenue. The economics of shaping are outstanding.

Opportunity 3: Capacity Market Tailwind. PJM capacity prices hit $666/MW-day (22x prior year). This adds ~$75/kW-yr to revenue for any resource that clears the capacity auction. Solar+storage can clear. This is an additional ~$8–10/MWh of revenue on top of energy + PPA premium.

Opportunity 4: Consequential Buyers Need Volume. Even if Meta/Amazon use consequential accounting, they still need to buy VRE — just the cheapest possible. 10+ GW of new wind/solar demand from just these two companies. For developers in MISO/SPP with low costs, this is a reliable floor of demand.

Opportunity 5: Behind-the-Meter Co-location. At least 46 data centers with 56 GW combined are already using behind-the-meter gas. Solar+storage co-located with data centers avoids interconnection queues, gets guaranteed offtake, and provides the DC with resilience benefits. This market is embryonic but enormous.

7. The Bottom Line: Where to Invest

Investment Decision Matrix: ISO × Product × Accounting Regime

Author analysis. Bubble size = addressable market (GW). Color = investment rating.

Tier 1 (Invest now): PJM solar+storage, ERCOT solar+storage, PJM wind
Tier 2 (Invest selectively): MISO solar+storage (if consequential survives), NYISO solar+storage (premium, small), CAISO solar+storage (selective)
Tier 3 (Wait/Avoid): Standalone solar anywhere cannibalization is active. MISO/SPP wind (oversupplied). CAISO standalone solar (unbankable).

The hyperscaler sustainability hedge in one sentence:

Hyperscaler data center demand creates a PPA price floor 30–140% above cannibalized wholesale value, making VRE development profitable even as merchant revenues collapse — but only for storage-paired products in ISOs where data centers are physically located, and only as long as at least some hyperscalers maintain credible climate commitments.

The durability of this hedge depends on three things: (1) the GHG Protocol outcome, (2) whether hyperscalers keep spending on clean energy as AI capex consumes budgets, and (3) whether storage pairing can keep capture rates above bankability thresholds. All three are currently favorable. The window for VRE developers to lock in premium PPAs is open now but narrowing as cannibalization accelerates and policy uncertainty grows.

8. Data Center Demand by ISO

Data Center Load: Existing vs. Pipeline by ISO (GW)

Sources: PJM 2025 Load Forecast; ERCOT LFL Reports; S&P Global 451 Research; EPRI Powering Intelligence 2026

ISO	EXISTING (GW)	UNDER CONSTRUCTION	COMMITTED/ PERMITTED	ANNOUNCED/ INTENDED	TOTAL PIPELINE	2030 FORECAST
PJM	14.1	6.5	11.0	25.0	56.6	35–44 GW
ERCOT	6.5	3.5	7.5	16.0	33.5	25–30 GW
CAISO	3.5	0.8	1.5	3.5	9.3	7–10 GW
MISO	2.5	1.5	3.5	7.5	15.0	7–12 GW
SPP	0.5	0.2	0.5	2.0	3.2	2–4 GW
NYISO	2.3	0.6	1.3	3.0	7.2	5–7 GW
NEISO	0.7	0.2	0.3	0.8	2.0	1.5–2 GW
US Total	**30.1**	**13.3**	**25.6**	**57.8**	**126.8**	**82–109 GW**

Note on materialization rates: *Sightline Climate estimates 30–50% of announced DC capacity will be delayed or cancelled. PJM assumes 35% materialization for planning. EPRI's low scenario assumes most under-construction + 25% of advanced planning. Apply a 35–50% haircut to "Announced/ Intended" for realistic projections.*

9. The Supply-Demand Gap

Hyperscalers have contracted ~80 GW of clean energy nameplate globally — but **nameplate ≠ firm-equivalent**. After applying capacity factors (~25–30% for solar/wind VPPAs), the firm-equivalent supply is only ~20–30 GW. Against 30+ GW of existing DC load and 80–110 GW in the pipeline, the gap is massive.

Clean Energy Supply vs. Data Center Demand by ISO (GW firm-equivalent)

Sources: Company sustainability reports; LBNL Queued Up 2024; PJM/ERCOT/MISO load forecasts; author synthesis of firm-equivalent calculations

ISO	NUCLEAR (FIRM GW)	VPPAS (NAMEPLATE GW)	VPPAS (FIRM-EQ GW)	SMR PIPELINE (GW)	QUEUE CLEAN (REALISTIC GW)	DC LOAD 2025 (GW)	DC LOAD 2030E (GW)	FIRM GAP 2030 (GW)
PJM	2.76	15–20	4.5–6.0	1.64	~62	14.1	35–44	**25–35**
ERCOT	0	10–15	3.0–4.5	0	~29	6.5	25–30	**20–27**
MISO	1.12	5–8	1.5–2.4	0	~63	2.5	7–12	**3–8**
CAISO	0	3–5	0.9–1.5	0	~65	3.5	7–10	**5–9**
NYISO	0	Minimal	<0.5	0	~15	2.3	5–7	**4–7**
NEISO	0*	Minimal	<0.5	0	~5	0.7	1.5–2	**1–2**
SPP	0	2–4	0.6–1.2	0	~17	0.5	2–4	**1–3**
US Total	**3.88**	**35–52**	**11–16**	**8.5–12**	**~256**	**30.1**	**82–109**	**59–91**

** NEISO has 3.3 GW of existing nuclear (Millstone + Seabrook) but none contracted to data centers. SMR pipeline figures are highly speculative — no SMR is under construction in the US. Queue clean figures apply ~13–27% completion rates (LBNL) to total clean energy in interconnection queues.*

The firm clean energy gap is 59–91 GW by 2030. Even including all SMR pipeline capacity (none under construction), the US needs ~50–80 GW of new firm clean energy for data centers alone. VPPAs at nameplate look impressive; at firm-equivalent, they cover only 15–20% of projected demand. This gap is the VRE developer's opportunity — especially for storage-paired products that can deliver firm-equivalent clean energy.

10. The Emissions Credibility Gap

The disconnect between hyperscaler sustainability claims and actual grid-level emissions is the elephant in the room. Market-based accounting allows companies to claim declining emissions while their physical electricity consumption — and the grid emissions it causes — are rising dramatically.

Location-Based Scope 2 Emissions: The Reality Behind the Claims ($MtCO_2$)

Sources: Google Sustainability Report 2024; Microsoft Sustainability Report 2024; Amazon Sustainability Report 2024

COMPANY	SCOPE 2 (LOCATION) 2020	SCOPE 2 (LOCATION) 2024	CHANGE	MARKET-BASED CLAIM	REALITY
Google	5.8 $MtCO_2$	11.2 $MtCO_2$	**+93%**	12% DC emission reduction (market-based)	Location-based emissions doubled. DC electricity up 27% YoY.
Microsoft	4.3 $MtCO_2$	~10.0 $MtCO_2$	**+133%**	Scope 1+2 down 30% (market-based)	Location-based more than doubled. TMI won't deliver until 2028.
Amazon	—	68.3 $MtCO_2e$	**+34% vs 2019**	100% renewable match (annual, 2023)	Emissions tripled since Climate Pledge. 3.8 GW clean added in 2025.
Meta	—	—	Rising significantly	100% renewable (annual)	Building 3+ GW of new gas while claiming clean energy leadership.
Apple	—	—	**-55% CO_2e**	100% renewable since 2014	Only major tech company with declining operational emissions.

Behind-the-Meter Gas: The Hidden Buildout

While sustainability reports tout renewable procurement, a parallel ~12+ GW behind-the-meter gas buildout is underway. These emissions are often untracked or unreported under current accounting frameworks.

Confirmed & Planned Behind-the-Meter Gas Capacity by Company (MW)

Sources: SELC; Data Center Dynamics; Bloomberg; TechCrunch; company filings

The credibility paradox: The companies most aggressively building gas (Meta: 3.2+ GW, Oracle: 4.3+ GW, xAI: 2.0 GW) are either claiming 100% renewable status or making no sustainability claims at all. The honest middle ground — transparently reporting location-based emissions and investing in firm clean energy — is occupied only by Google and, increasingly, Microsoft. For VRE developers, this means: **the most reliable PPA buyers are the ones with the most credible (and therefore most demanding) accounting frameworks.**

11. Technology Cost Trajectories & Learning Curves

Cost decline rates determine which resources will be competitive for hyperscaler PPAs by 2030–2035. Solar and batteries are on steep learning curves; nuclear and geothermal have potential but unproven trajectories at scale.

LCOE Trajectories by Technology ($/MWh, Unsubsidized Midpoint)

Sources: Lazard LCOE+ v18 (June 2025); NREL ATB 2024 Moderate scenario; Fervo Energy; Form Energy; Carbon Direct

TECHNOLOGY	2025 $/MWH	2030E	2035E	LEARNING RATE	OBBBA IMPACT
Utility Solar	$38–78	$28–50	$22–40	**20–24%**	60% credit '26 → 0% by '28
Onshore Wind	$37–86	$30–60	$27–55	10–15%	Same phase-down as solar
Battery 4hr (LCOS)	$78–254	$55–150	$40–100	**35%**	Preserved through 2032
LDES Iron-Air	$150–250	$80–120	$50–80	~15% (modeled)	Preserved (storage ITC)
Geothermal EGS	$60–100	$45–70	$35–55	15–20%	Preserved through 2032
Nuclear SMR	—	$100–150 (FOAK)	$80–120	0% (historical)	45U PTC preserved through 2032
CCS-CCGT (net of 45Q)	$45–75	$40–65	$35–60	5–10%	**45Q expanded ($85/ton)**
Solar+4hr Storage	$50–131	$40–90	$30–70	—	Blended: solar cliff + storage preserved

The cost crossover: By 2030, solar+4hr storage ($40–90/MWh) will be cheaper than new gas CCGT ($48–109/MWh) in most ISOs. Enhanced geothermal ($45–70/MWh) becomes competitive with existing nuclear PPAs ($40–60/MWh). Iron-air LDES ($80–120/MWh) crosses below nuclear SMR FOAK ($100–150/MWh). **For VRE developers: the shaped product (solar+storage) is already cost-competitive with fossil alternatives — the green premium is pure upside.**

12. The Hourly vs. Annual Technology Demand Split

The accounting framework determines *which* technologies get built. Hourly matching is the sole demand driver for LDES, geothermal, and new nuclear. Annual matching drives solar/wind volume but provides zero market signal for firm clean technologies.

Resource Demand by Procurement Strategy (GW by 2035)

Sources: Princeton ZERO Lab (Riepin et al. 2025); RMI Clean Power by the Hour; NREL ATB 2024; author synthesis

CFE TARGET	COST PREMIUM VS. ANNUAL	APPROX $/MWH ADDER	KEY TECHNOLOGY UNLOCKED
70–80%	~0%	$0	Solar + wind + grid baseline
80–90%	~5–15%	$3–8	+ 4hr battery storage
90–95%	~15–30%	$8–18	+ LDES or clean firm
98%	**~54%**	$25–35	+ Geothermal / nuclear / CCS
100% (with LDES+firm)	**~42–64%**	$25–40	Full portfolio: all of the above
100% (VRE+battery only)	**~200%+**	$60–100+	Massive overbuild. Impractical.

> **The 3% tipping point:** Princeton ZERO Lab found that if companies representing just **3% of C&I electricity demand** committed to 24/7 CFE, it would quadruple LDES deployment and drive 25% cost reduction in iron-air batteries by 2030. Google and Microsoft alone represent ~1.5% of US electricity. Add a few more hourly buyers and the entire firm clean technology sector tips into viability.

13. OBBBA Policy Landscape

The One Big Beautiful Bill Act (signed July 4, 2025) reshaped the clean energy incentive landscape. Solar and wind face a subsidy cliff; storage, geothermal, nuclear, and CCS are largely preserved.

IRA Credit Phase-Down Timeline by Technology

Sources: Akin Gump; Grant Thornton; Tax Foundation analysis of OBBBA enacted provisions

TECHNOLOGY	PRE-OBBBA	POST-OBBBA	NET IMPACT
Solar/Wind (PTC/ ITC)	Full credit through 2032+	**60% in 2026 → 20% in 2027 → 0% from 2028**	$336B investment cut est. (Aurora)
Battery Storage ITC	30% + bonuses through 2032	Preserved. Phase-down 2032–2036	Forecast grew 15% post-OBBBA
Geothermal (48E ITC)	Full ITC	Preserved. Same phase-down as storage	Minimal impact near-term
Nuclear PTC (45U)	Through 2032	Three-step phase-down after 2032	Existing fleet protected
45Q (CCS)	$85/ton for saline	**Preserved & expanded (EOR at parity)**	Most favorable outcome
Clean Hydrogen (45V)	Begin construction by 2033	**Terminated for construction after Dec 2027**	Green H_2 development halted

The solar/wind cliff matters for VRE developers: Projects must begin construction by July 2026 for full ITC/PTC credit. After 2028, unsubsidized solar ($38–78/MWh) is still competitive with gas ($48–109/MWh) but developer IRRs compress by 2–4 percentage points. The PPA green premium becomes even more critical as the only source of above-market returns. **Safe harbor now, lock in PPAs now.**

Sources & Citations

- LBNL Utility-Scale Solar, 2024/2025 Edition — Capture rates, value factors, PPA pricing
- LevelTen Energy PPA Price Index — Q2/Q3 2025 PPA pricing data
- BNEF Corporate Energy Market Outlook 2026 — Corporate PPA volumes, hyperscaler share
- PJM 2025 Long-Term Load Forecast — 30 GW DC growth by 2030
- Modo Energy: PJM Demand Forecast Analysis — Supply-constrained demand scenarios
- Latitude Media: ERCOT Large Load Queue — 233 GW queue, 70% data centers
- EPRI Powering Intelligence 2026 — 380–790 TWh DC demand by 2030
- S&P Global 451 Research — 134 GW DC demand by 2030
- Meta: Emissions First Partnership — Consequential accounting position
- Latitude Media: Big Tech Two Camps — Hourly vs. consequential split
- NewClimate Institute: 24/7 vs. EFP Briefing — GHG Protocol debate analysis
- TechCrunch: Meta Louisiana Gas Plants — 2.25 GW new gas for Meta DC
- Amperon: US Curtailment Exploding — 4 of 7 ISOs set curtailment records
- Sightline Climate: DC Pipeline Materialization — 30–50% attrition rate
- pv magazine: Solar+Storage PPA Structures — Hybrid PPA models
- Ember: How Cheap is Battery Storage? — 40% cost drop in 2024
- McKinsey: How Hyperscalers Fuel 24/7 Clean Power — Hyperscaler demand dynamics
- WattTime: Impact Accounting vs. Hourly Matching Review — 4.7x emissions avoided claim
- Synertics: Spain Solar Capture Rates — Spain cannibalization reference case
- Lazard LCOE+ v18 (June 2025) — Unsubsidized LCOE ranges by technology
- NREL Annual Technology Baseline 2024 — LCOE projections through 2050
- Princeton ZERO Lab — Riepin et al. 2025 (Joule) — 24/7 CFE system-level impacts; 3% tipping point
- RMI: Clean Power by the Hour — Hourly matching technology demand signals
- Fervo Energy — Enhanced geothermal cost trajectory; drilling time 150→15 days
- Carbon Direct: CCS for Gas Power — CCS-CCGT LCOE and 45Q economics
- Global CCS Institute: OBBBA 45Q — 45Q credit preservation and expansion
- Akin Gump: OBBBA Energy Tax Analysis — Solar/wind ITC/PTC phase-down schedule
- Deloitte: Nuclear and Data Centers — 35–65 GW nuclear projection over next decade
- pv magazine: Google/Form Energy 30 GWh Iron-Air — Landmark LDES deployment
- Google Sustainability Report 2024 — 66% global CFE; 27% electricity consumption growth
- Meta 2025 Sustainability Report — Nuclear pipeline, emissions trajectory

CSV Data Files

Raw data underlying this analysis is available in `data/dc-clean-energy-research/`:

- `dc_load_by_iso.csv` — Data center capacity by ISO, metro, and development stage
- `demand_forecasts_by_iso.csv` — Multi-source demand projections through 2040
- `hyperscaler_projects.csv` — Company-level capacity, climate targets, PPAs

- `capture_rates_by_iso.csv` — Solar/wind capture rates by ISO (2020–2035)
- `ppa_value_framework.csv` — PPA prices, wholesale values, green premiums
- `vre_investment_thesis.csv` — Investment ratings by ISO and product type
- `accounting_regime_scenarios.csv` — Hourly vs. consequential vs. hybrid scenario modeling
- `clean_energy_supply_by_iso.csv` — Nuclear PPAs, VPPAs, SMR pipeline, BESS, queue data by ISO
- `btm_gas_buildout.csv` — Behind-the-meter gas capacity by company and location
- `emissions_trajectory.csv` — Location vs. market-based Scope 2 emissions by company
- `dc_load_profile_parameters.csv` — Parameters for constructing 8,760-hour DC load profiles
- `resource_mix_projections.csv` — Resource demand by procurement strategy (hourly/annual/grid) through 2035
- `cost_projections.csv` — LCOE trajectories, learning rates, and OBBBA impacts by technology